institution_id+student_id
institution_id+user_id
institution_id+section_id
institution_id+section_id+course_id
institution_id+attempt_id+quiz_id+student_id+section_id
institution_id+actor_user_id
institution_id+course_id
institution_id+section_id+course_id
institution_id+actor_user_id
user_id
institution_id+course_id
institution_id+section_id+course_id
institution_id+quiz_id+course_id
institution_id+course_id
institution_id+created_by
institution_id+course_id
institution_id
institution_id+quiz_id+course_id
institution_id+created_by
institution_id+grade_item_id+quiz_id
institution_id+course_id
institution_id+course_id
institution_id+module_id
institution_id+created_by
institution_id+student_id

USERS			
uuid	id	PK	NOT NULL
varchar(254)	email		NOT NULL
varchar(255)	password_hash		NOT NULL
varchar(200)	name		NOT NULL
boolean	active		NOT NULL
timestamptz(6)	created_at		NOT NULL
timestamptz(6)	updated_at		NOT NULL

INSTITUTIONS			
uuid	id	PK	NOT NULL
varchar(63)	slug		NOT NULL
varchar(200)	name		NOT NULL
boolean	active		NOT NULL
timestamptz(6)	created_at		NOT NULL
timestamptz(6)	updated_at		NOT NULL

INSTITUTION_MEMBERSHIPS			
boolean	is_admin		NOT NULL DEFAULT false
boolean	is_super_admin		NOT NULL DEFAULT false
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	user_id	FK	NOT NULL
boolean	active		NOT NULL
timestamptz(6)	created_at		NOT NULL
timestamptz(6)	updated_at		NOT NULL

COURSES			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
varchar(32)	code		NOT NULL
varchar(200)	name		NOT NULL
varchar(16)	term		NOT NULL
uuid	created_by	FK	NOT NULL
timestamptz(6)	created_at		NOT NULL
timestamptz(6)	grade_items_locked_at		NULL monotono

SECTIONS			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	course_id	FK	NOT NULL
varchar(32)	code		NOT NULL
integer	capacity		NULL
timestamptz(6)	created_at		NOT NULL

GRADE_PUBLICATIONS			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	course_id	FK	NOT NULL
uuid	actor_user_id	FK	NOT NULL
text	idempotency_key		NOT NULL 1..200
jsonb	request		NOT NULL privado
integer	published_count		NOT NULL positivo
timestamptz(6)	published_at		NOT NULL

COURSE_MODULES			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	course_id	FK	NOT NULL
varchar(200)	title		NOT NULL
integer	position		NOT NULL
timestamptz(6)	published_at		NULL

ENROLLMENTS			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	section_id	FK	NOT NULL
uuid	user_id	FK	NOT NULL
varchar(16)	role		NOT NULL
boolean	active		NOT NULL
timestamptz(6)	created_at		NOT NULL

AUDIT_EVENTS			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	actor_user_id	FK	NULL solo migracion operativa auditada
varchar(64)	action		NOT NULL
varchar(32)	resource_type		NOT NULL
uuid	resource_id		NOT NULL; referencia polimorfica sin FK
uuid	course_id	FK	NULL solo evento institucional
uuid	section_id	FK	NULL
timestamptz(6)	occurred_at		NOT NULL
jsonb	changes		NOT NULL

QUIZZES			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	course_id	FK	NOT NULL
uuid	section_id	FK	NULL
varchar(200)	title		NOT NULL
varchar(10000)	instructions		NULL
timestamptz(6)	opens_at		NULL
timestamptz(6)	closes_at		NULL
integer	max_attempts		NULL
timestamptz(6)	published_at		NULL
uuid	created_by	FK	NOT NULL
jsonb	questions		NOT NULL

MATERIALS			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	module_id	FK	NOT NULL
varchar(200)	title		NOT NULL
varchar(16)	kind		NOT NULL markdown
text	markdown_body		NOT NULL 1..10000
timestamptz(6)	published_at		NULL
uuid	created_by	FK	NOT NULL

QUIZ_ATTEMPTS			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	course_id	FK	NOT NULL
uuid	quiz_id	FK	NOT NULL
uuid	section_id	FK	NOT NULL
uuid	student_id	FK	NOT NULL
integer	attempt_number		NOT NULL
varchar(16)	status		NOT NULL
timestamptz(6)	started_at		NOT NULL
timestamptz(6)	submitted_at		NULL
timestamptz(6)	cancelled_at		NULL
jsonb	answers		NOT NULL
numeric(8,2)	score_points		NULL
numeric(5,2)	score_percent		NULL
numeric(3,1)	grade_value		NULL
text	start_key		NOT NULL 1..200
text	submit_key		NULL 1..200

GRADE_ITEMS			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	course_id	FK	NOT NULL
uuid	quiz_id	FK	NOT NULL
varchar(200)	title		NOT NULL
numeric(5,2)	weight_percent		NOT NULL
timestamptz(6)	created_at		NOT NULL

GRADES			
uuid	id	PK	NOT NULL
uuid	institution_id	FK	NOT NULL
uuid	grade_item_id	FK	NOT NULL
uuid	quiz_id	FK	NOT NULL
uuid	student_id	FK	NOT NULL
uuid	section_id	FK	NOT NULL
uuid	attempt_id	FK	NOT NULL
numeric(5,2)	score_percent		NOT NULL
numeric(3,1)	grade_value		NOT NULL
timestamptz(6)	published_at		NULL
timestamptz(6)	created_at		NOT NULL
timestamptz(6)	updated_at		NOT NULL